Learning Double Subspace Representation for Joint Hyperspectral Anomaly Detection and Noise Removal

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***Abstract*— Efforts to enhance the detection accuracy of hyperspectral (HS) anomaly detection (AD) have been significant, but the impact of noise resulting from HS data acquisition and transmission has not been well studied. Furthermore, the separation of denoising and subsequent interpretation makes it challenging to evaluate and control the influence of noise on the detection results. To this end, we proposed a joint AD and noise removal (ADNR) paradigm called DSR-ADNR, which develops a double subspace representation method to obtain both denoised and detection results simultaneously. DSR-ADNR uses a low-dimensional orthogonal basis to represent HS images and extract distinctive features for AD. The feature matrix is represented by a dictionary-based low-rank subspace that captures the complex nature of the low-dimensional features. In each iteration, DSR-ADNR utilizes the nonlocal self-similarity of the feature matrix to remove noise and improve intermediate detection performance. Meanwhile, the progressive LR representation of the background and anomalies for the feature matrix upgrades the explicit LR expression of nonlocal self-similar patches for better denoising. The well-designed linearized alternating direction method of multipliers with an adaptive penalty (LADMAP) is utilized to solve the proposed DSR-ADNR. Extensive experiments on simulated and real-world datasets demonstrate the effectiveness of DSR-ADNR in the HS AD task under different noise cases.**

***Index Terms*— Anomaly detection (AD), linearized alternating direction method of multipliers with an adaptive penalty (LADMAP), noise removal, subspace representation.**

I. Introduction

OWING to high spectral resolution, hyperspectral (HS) imaging techniques play an increasingly significant role in Earth observation missions with a wide range of applications, e.g., unmixing [1], [2], [3], classification [4], [5], and target detection [6], [7]. Anomaly detection (AD) is one such application that seeks to distinguish abnormal objects in the image scene. The potential anomalies of interest possess distinct spectral characteristics and occur in low-probabilities areas of the image.

Research on AD mainly focuses on the distinction between background and anomalies since no prior knowledge of anomalies is available. Reed and Xu [8] proposed the classical Reed–Xiaoli (RX) algorithm early in 1990, which was well-known as one representative statistical AD baseline. This detector assumes that the background follows a Gaussian distribution, and it computed the mean vector and covariance matrix of input HS images to obtain the Mahalanobis distance between a testing pixel and the surroundings. Several extensions based on RX have been reported for AD, including regularized RX [9], local RX [10], subspace RX (SSRX) [11], nonlinear kernel RX (KRX) [12], and improved KRX [13]. Local RX exploited local statistics to estimate the background but ignored the complexity of real HS data [10]. SSRX was presented by using the principal component analysis (PCA) of the background covariance matrix [11]. A nonlinear KRX method modeled an HS image in a feature space [12]. An improved KRX was developed to overcome the rank deficiency of the basic KRX detector [13]. A cluster KRX (CKRX) detector grouped background pixels into clusters and then used a fast eigendecomposition algorithm to generate the AD index. CKRX had comparable detection performance as KRX but with much lower computational requirements [14].

In addition to RX and its variants, various state-of-the-art reconstruction-based AD methods have been deployed, such as collaborative representation (CR), sparse representation (SR), and low-rank representation (LRR). Nevertheless, the exact representation coefficients are shared among all spectral signatures in the basic CRD version. To alleviate this problem, Wu et al. [15] advocated a novel relaxed CR detector for HS AD via a well-designed nonglobal dictionary. Feng et al. [16] combined total variation (TV) regularized low-rank tensor decomposition with CR. Ling et al. [17] employed SR to approximately represent the background as a sparse linear combination of its spatial surrounding pixels. Zhu et al. [18] presented a single-spectrum-driven binary-class SR target detector (SSBSTD) via target and background dictionary construction. Inspired by homogeneous region-based Joint

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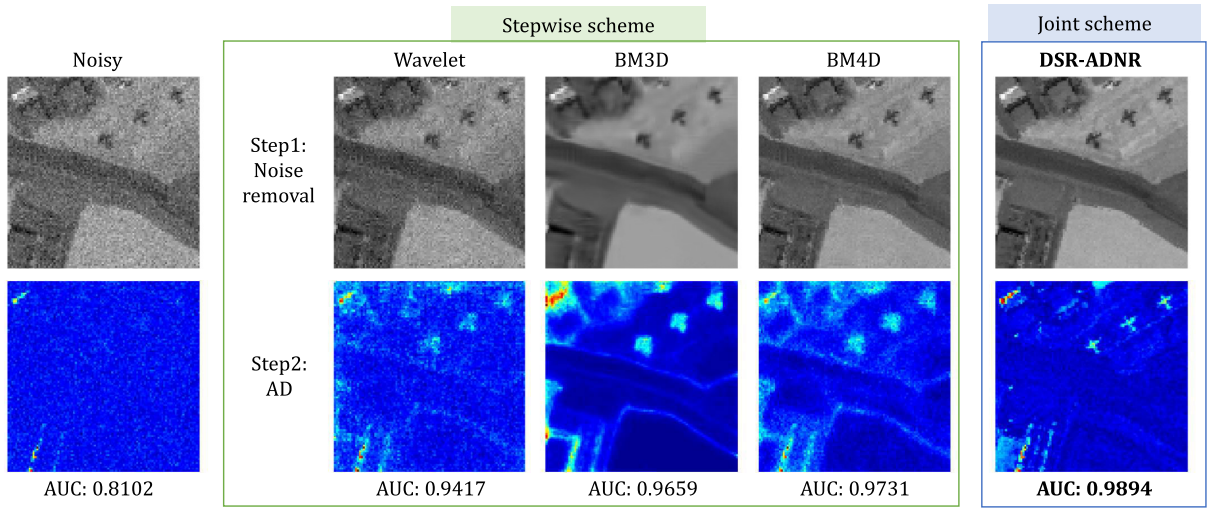


Fig. 1. Denoised images and AD results by different methods.

SR, Yang et al. [19] proposed a robust background feature extraction method. However, the above SR and CR approaches meet the bottleneck in detection performances due to their pixel-by-pixel detection [20].

LRR was first proposed in [21], which proved LRR enables potential subspace representation even when observations are contaminated by outliers. Due to spatial homogeneity and spectral correlation, the background has global LR characteristics. Anomalies are sparse because of their small proportion and low probability of occurrence. Based on the LRR theory, an HS image is decomposed as the background and anomalies. The robust PCA (RPCA) [22], [23] algorithm enables the division of two-dimensional HS data into the LR and SR (LRASR) parts [24]. Consequently, the LR and sparse matrix decomposition (LRaSMD) model was proposed for HS AD, and the go decomposition algorithm was adopted to solve the HS AD optimization [25]. Niu and Wang [26] proposed a novel HS anomaly detector based on LRR and a learned dictionary, obtaining a background dictionary via sparse dictionary learning. However, this detector ignores abnormal pollution during dictionary learning. Xu et al. [27] injected a sparsity-inducing regularization into the LRR formulation and proposed an LRASR method. Besides the background dictionary, Huyan et al. [28] explored the prior information of anomalies hidden in the scene to construct a potential anomaly dictionary. In [27] and [28], LR and l_1 sparsity-inducing regularizations are used to describe the local structure of the background. The sparsity criterion assumes that each pixel covers a limited number of materials, while, for the low rankness criterion, the number of distinct materials in a scene is much less than the number of bands. However, these two criteria may be redundant to some extent in their physical meaning. Cheng and Wang [29] incorporated the graph and TV regularization into the LRR optimization (GTVLRR). Feng et al. [30] advocated an anomaly detector for HS images based on LRaSMD with density peak-guided CR. Zhao et al. [31] utilized a novel enhanced TV with an endmember background dictionary (EBD) for HS AD, which decreased the impact of anomaly components in mixed pixels. With the advent of deep learning, Xie et al. [32] utilized a weakly supervised LRR to formulate the background. Cheng and Wang [33] fused TV and sparsity-inducing decomposition regularizations to facilitate the separation. Wang et al. [34] combined LR prior with autoencoders to design a deep LR prior-based AD approach. Zhao et al. [35] proposed a spectral–spatial HS AD method based on CR constraint stacked autoencoders. Inspired by the high-dimensional structure of HS images, Li et al. [36] employed the Tucker decomposition and sparse tensor representation for HS AD. Wang et al. [37] developed a novel tensor LR method with a PCA preprocessing step, which was the first time to introduce tensor LRR into HS AD.

Real-world noise typically exists in all HS data due to the limitations of imaging systems and equipment conditions [38], [39], [40]. Qu et al. [41] presented a two-step approach for AD in HS images contaminated by noise, entitling abundance- and dictionary-based LR (ADLR) decomposition. The abundance matrix is extracted from the noisy images, followed by the extraction of anomalies from the abundance matrix via a low-rank decomposition. Three differences between ADLR and our method can be noted.

1) ADLR belongs to a two-step process, while our proposed method is a joint scheme. The denoised and detection results would be simultaneously produced by our method.
2) The preprocessing unmixing has a significant impact on AD. The detection performance may be worse when the results of unmixing are unsatisfied.
3) ADLR focuses on unmixing and AD, whereas we concentrate on denoising and AD. Therefore, we conduct extensive experiments with different noise levels in Section III.

Noise has a significant negative impact on AD in HS images since it contaminates the original geometrical structure of the HS image background and anomalies. However, recent research on AD is hardly absorbed in noisy scenes and carried out on denoising datasets. Coupled noise removal and AD pose a novel challenge in the HS remote sensing (RS)

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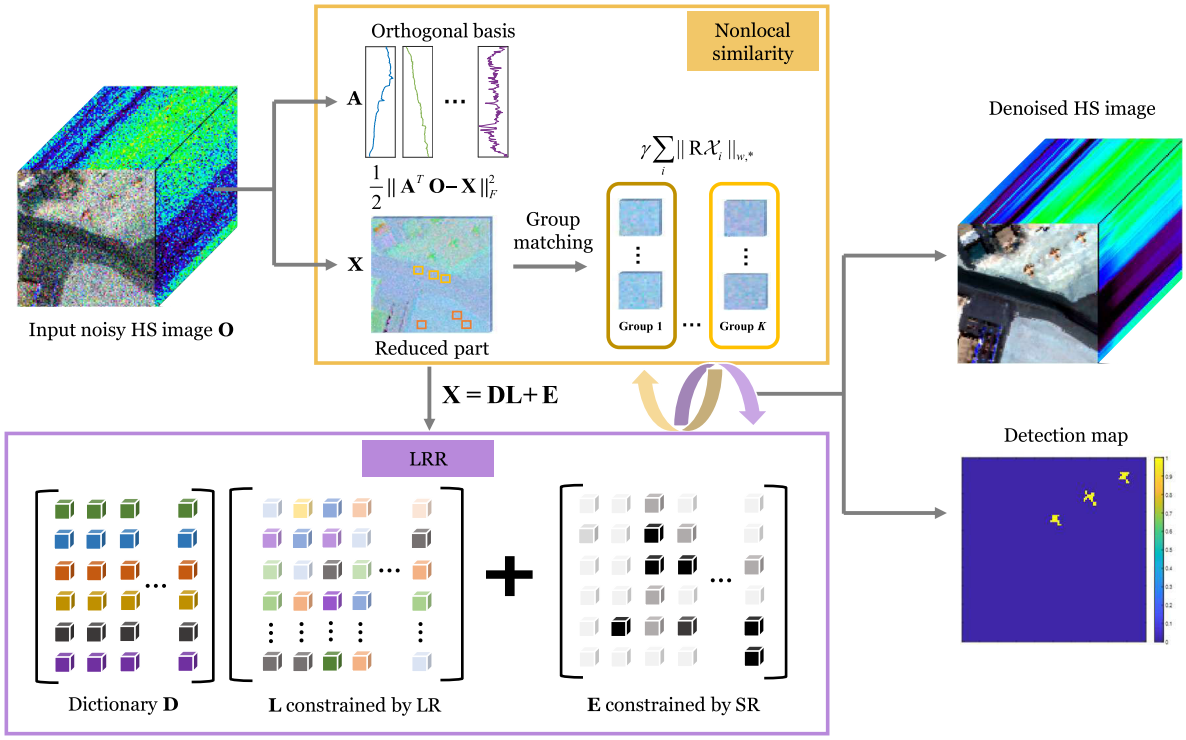


Fig. 2. Schematic of the proposed DSR-SDNR model.

community. Methodology and experiments that account for real-world problems are necessary to obtain more meaningful results. In this article, we place particular emphasis on the coordination between noise removal and AD, which provides one novel research hotpot for intelligent HS RS development. A joint mechanism is proposed to accomplish data processing and interpretation tasks simultaneously. Noise removal and AD boost each other in one united iteration loop. The main contributions of this study are summarized as follows.

1) We propose a unified AD and noise removal (ADNR) paradigm in an innovative manner, which jointly learns double subspace representation and alternatively updates processing and interpretation tasks. The integration of joint preprocessing and interpretation enables us to address practical problems more effectively and promote the future development of HS RS intelligence.
2) Rather than using the original raw pixels, we lie an HS image in a spectral subspace with a low-dimensional orthogonal basis. This approach benefits the feature refinement of AD and reduces computational time. Assist by the underlying subspace of LRR, AD also contributes to better denoised results by enhancing the explicit expression of nonlocal self-similarity.
3) We devise a novel double subspace representation strategy called DSR-ADNR, which alternates between updating ADNR. To solve this model more effectively, we developed a linearized alternating direction method of multipliers with an adaptive penalty (LADMAP).

The remaining sections of this article are organized as follows. Section II introduces the study of the proposed model. Some experimental results and analysis are shown and discussed in Section III. Conclusions and future work are reported in Section IV.

II. Proposed Method

A. Motivation

HS images are inevitably corrupted by mixed noises during data acquisition and transmission, making it challenging to accurately observe and recognize the materials on the surface of the Earth [42], [43], [44]. Generally, a common strategy is to remove the noise first, followed by the AD procedure. Nevertheless, the inherent relationship between HS noise removal and AD is severed in this stepwise scheme. Since denoising and interpretation are typically separated, it is difficult to manage and quantify the influence of denoising on AD. On the one hand, the traditional two-step approach can result in a loss of image detail and compromise the accuracy of the AD procedure. On the other hand, insufficient denoising can negatively affect the performance of AD. Here, we select three denoising methods and the global RX detector to compare the stepwise approach with our proposed joint scheme. A benchmark San Diego dataset was contaminated with zero-mean additive Gaussian noise with a variance of 20/255. As shown in Fig. 1, Wavelet [45], BM3D [46], and BM4D [47] are used to remove the Gaussian noise. In the second row of Fig. 1 (a)–(d), AD maps of noisy and denoised HS images are obtained by the RX detector. After denoising, all AUC values are improved to different degrees. Some residual Gaussian noise still exists in the denoised result of Wavelet, which limited the accuracy of anomaly locations. BM3D and BM4D achieve noise removal better than

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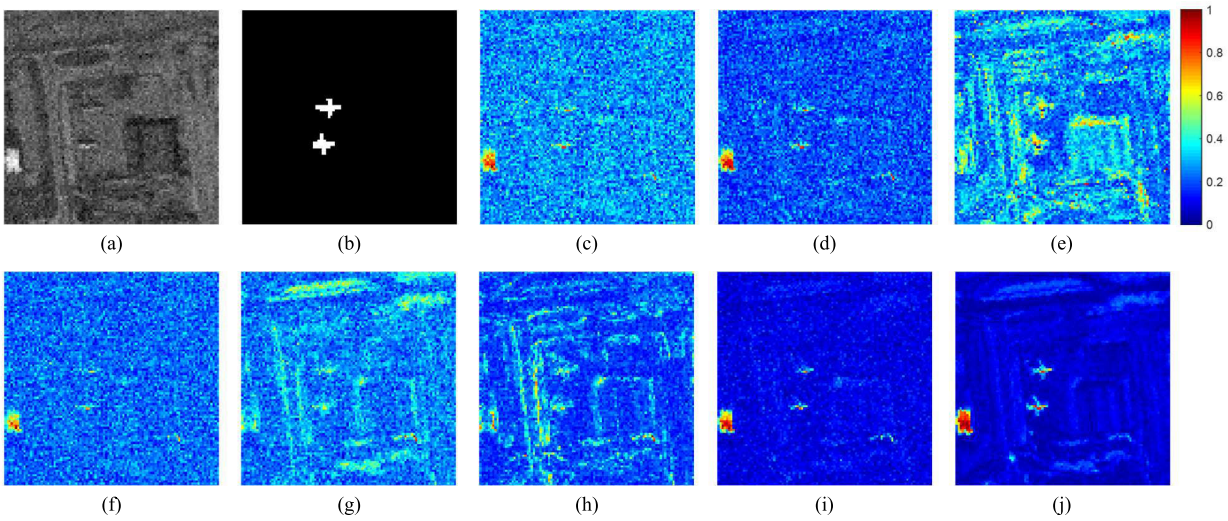


Fig. 3. Detection results of all compared methods: (a) noisy Airport image under case 2, (b) reference map, (c) global RX, (d) DWT RX, (e) CKRX, (f) RPCA, (g) LRSAR, (h) GTVLRR, (i) LSMAD, and (j) DSR-ADNR.

Wavelet, while high false alarm rates (FARs) appear on the edges of three flights. Especially, BM3D losses more structure information than BM4D and, thus, obtains a lower AUC value than BM4D. Our joint scheme performs better than the other two-step scheme and even gains the best AUC score. These results illustrate the motivation for combining noise removal and AD for HS imagery.

B. DSR-ADNR Model

The overview of the proposed paradigm is shown in Fig. 2. One HS image $\mathbf{O}$ is input under a noisy case. This image lies in a spectral subspace based on an orthogonal basis $\mathbf{A}$, producing a reduced part $\mathbf{X}$ with distinct feature extraction. The low-dimensional feature matrix $\mathbf{X}$ can be divided into two parts: the background and anomalies $\mathbf{E}$. The underlying subspace of LRR is explored to represent the background as $\mathbf{DL}$. Noise removal boosts AD in one united iteration loop. The identification of outliers is also profitable for the seeking of similar image patches. With mutual reinforcement, a denoised HS image and a detection map are output concurrently.

Assuming that an HS image $\mathcal{O} \in \mathbb{R}^{h\times v\times z}$ is corrupted by noise, the simultaneous noise removal and AD task in third unfolding forms can be formulated as follows:

$$\begin{cases} \text{Noise removal:} & \mathbf{O} = \mathbf{H} + \mathbf{N} \\ \text{AD:} & \mathbf{X} = \mathbf{DL} + \mathbf{E} \end{cases}$$

where $\mathbf{H} = \mathbf{AX}$ denotes the latent clean HS image, and $\mathbf{O}$, $\mathbf{N}$, $\mathbf{E} \in \mathbb{R}^{z\times \mathrm{hv}}$, $\mathbf{A} \in \mathbb{R}^{z\times k}$, $\mathbf{X} \in \mathbb{R}^{k\times \mathrm{hv}}$, $\mathbf{D} \in \mathbb{R}^{k\times d}$, and $\mathbf{L} \in \mathbb{R}^{d\times \mathrm{hv}}$ are the noisy HS image, additive noise, sparse part, the orthogonal basis, the reduced component, the background dictionary, and representation coefficients, respectively. We consider the decomposition of $\mathbf{O}$ with the orthogonal basis $\mathbf{A}$ and the LR of $\mathbf{X}$, proposing a comprehensively joint function as follows:

$$\min_{\mathbf{X},\mathbf{A},\mathbf{E},\mathbf{L}} \frac{1}{2}||\mathbf{O} - \mathbf{AX}||_F^2 \quad \text{s.t.} \quad \mathbf{A}^{\mathrm{T}}\mathbf{A} = \mathbf{I}, \quad \mathbf{X} = \mathbf{DL} + \mathbf{E} \tag{1}$$

where ith column of $\mathbf{A}$ corresponds to the ith signature of an HS image. The third dimension of the corresponding coefficient part $\mathcal{X}$ can be regarded as the feature map.

Remark 1: The significance of the orthogonal constraint $\mathbf{A}^{\mathrm{T}}\mathbf{A} = \mathbf{I}$ is illustrated here. On the one hand, the orthogonality of $\mathbf{A}$ boosts each base to be more obvious from each other, conducing to noise out of $\mathbf{A}$ and further a closed-form solution of the $\mathbf{A}$-related subproblem. The distribution of noise allows the estimation of the remaining noise levels in the reduced part. On the other hand, $\mathbf{A}^{\mathrm{T}}\mathbf{O}$ is regarded as the distinct feature extraction of original images $\mathbf{O}$, which urges the application of LRR to separate the background and anomalies.

Via the orthogonal basis $\mathbf{A}$, this spectral subspace also contributes to the feature extraction of AD. The principal components of LR background are refined by a conducted dictionary $\mathbf{D}$, e.g., $\mathbf{X} = \mathbf{DL} + \mathbf{E}$. The strategy for dictionary construction in [27] is adopted here. Pixels of $\mathbf{X}$ are divided into several clusters by k-means. Each dictionary atom has the same first-mode dimension of k as the pixel of $\mathbf{X}$, where k is much smaller than the first-mode dimension z of the original noisy image $\mathbf{O}$. The number of dictionary atoms d is empirically discussed in Section III-A. The background dictionary consists of the elements determined by the basic RX algorithm, which have lower detection results than the others in each cluster.

A local patch always has many similar patches across an HS image since this HS image frequently contains many repetitive local spatial patterns [48]. Instead of the original HS image, a reduced image is segmented into 3-D full-band patches (FBPs) to decrease the time cost. All nonlocal similar FBPs are stacked and converted as matrices with their spatial mode and spectral mode. With the nonlocal similarity of FBPs, (1) is rewritten as

$$\min_{\mathbf{X},\mathbf{A},\mathbf{E},\mathbf{L}} \frac{1}{2}||\mathbf{O} - \mathbf{AX}||_F^2 + \gamma \sum_i ||\mathrm{R}_i\mathcal{X}||_{w,*} \quad \text{s.t.} \quad \mathbf{A}^{\mathrm{T}}\mathbf{A} = \mathbf{I}, \quad \mathbf{X} = \mathbf{DL} + \mathbf{E}. \tag{2}$$

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$\mathrm{R}_i\mathcal{X} \in \mathbb{R}^{p^2\times k\times q}$ represents the unfolding matrix of $\Re_i\mathcal{X}$ along model-2. The operation of $\Re_i\mathcal{X}$ contains two steps: 1) segment the tensor $\mathcal{X}$ as an overlapped patch tensor $\mathcal{X}_i \in \mathbb{R}^{p\times p\times k}$ and 2) cluster w similar patches in a neighborhood area by computing Euclidean distance. $||\cdot||_{w,*}$ denotes the weighted nuclear norm minimization (WNNM) [49].

Each pure endmember in HS imagery primarily corresponds to one material. Nevertheless, many mixed pixels exist in a single image due to the stratified distribution of objects. According to the LRR, the underlying HS data are drawn from a union of subspaces. The resulting optimization can be expressed as

$$\min_{\mathbf{X},\mathbf{A},\mathbf{E},\mathbf{L}} \frac{1}{2}||\mathbf{O}-\mathbf{A}\mathbf{X}||_F^2+\gamma\sum_i||\mathrm{R}_i\mathcal{X}||_{w,*}+\beta||\mathbf{L}||_*+\lambda||\mathbf{E}||_1 \quad \text{s.t.}\quad \mathbf{A}^{\mathrm{T}}\mathbf{A}=\mathbf{I},\quad \mathbf{X}=\mathbf{D}\mathbf{L}+\mathbf{E} \tag{3}$$

where γ, β, and λ are three regularization parameters to control the tradeoff among four terms. The first term $||\mathbf{A}^{\mathrm{T}}\mathbf{O}-\mathbf{X}||_F^2$ is the data fidelity terms to decompose $\mathbf{O}$ into the spectral LR orthogonal basis $\mathbf{A}$ and the feature matrix $\mathbf{X}$. The second term $||\mathrm{R}_i\mathcal{X}||_{w,*}$ corresponds to the WNNM of $\mathbf{X}$ to constrain its nonlocal self-similarity. The third term $||\mathbf{L}||_*$ is used to characterize the LR property of representation coefficient $\mathbf{L}$. The sparse term $||\mathbf{E}||_1$ is a l_1 norm to find the anomaly locations.

C. Optimization Procedure

To solve the above joint noise removal and AD problem, we introduce one auxiliary variable $\mathcal{W}$. Equation (3) is reformulated as

$$\min_{\mathbf{X},\mathcal{W},\mathbf{A},\mathbf{E},\mathbf{L}} \frac{1}{2}||\mathbf{O}-\mathbf{A}\mathbf{X}||_F^2+\gamma\sum_i||\mathrm{R}_i\mathcal{W}||_{w,*}+\beta||\mathbf{L}||_*+\lambda||\mathbf{E}||_1 \quad \text{s.t. } \mathbf{A}^{\mathrm{T}}\mathbf{A}=\mathbf{I},\quad \mathbf{X}=\mathbf{D}\mathbf{L}+\mathbf{E},\quad \mathcal{X}=\mathcal{W}. \tag{4}$$

Based on LADMAP [50], (4) is transformed into minimizing the following augmented Lagrangian function:

$$\begin{aligned}\min_{\mathbf{X},\mathcal{W},\mathbf{A},\mathbf{E},\mathbf{L}} &\frac{1}{2}||\mathbf{O}-\mathbf{A}\mathbf{X}||_F^2+\gamma\sum_i||\mathrm{R}_i\mathcal{W}||_{w,*}\\ &+\beta||\mathbf{L}||_*+\lambda||\mathbf{E}||_1+\langle\mathbf{Y}_1,\mathbf{X}-\mathbf{D}\mathbf{L}-\mathbf{E}\rangle\\ &+\langle\mathcal{Y}_2,\mathcal{X}-\mathcal{W}\rangle\\ &+\frac{\mu}{2}\left(||\mathbf{X}-\mathbf{D}\mathbf{L}-\mathbf{E}||_F^2+||\mathcal{X}-\mathcal{W}||_F^2\right)\end{aligned} \tag{5}$$

under the constraint $\mathbf{A}^{\mathrm{T}}\mathbf{A}=\mathbf{I}$, where μ is the penalty parameter. $\mathbf{Y}_1$ and $\mathcal{Y}_2$ are two multipliers used in the LADMAP iterations. Herein, (5) is optimized alternatively over one variable while fixing the others. In the tth iteration, variables involved in (5) is updated as follows.

1) *Update* $\mathbf{A}_t$*:* The subproblem of $\mathbf{A}_t$ is given as

$$\mathbf{A}_t=\underset{\mathbf{A}^{\mathrm{T}}\mathbf{A}=\mathbf{I}}{\operatorname{argmin}}\frac{1}{2}||\mathbf{O}_t-\mathbf{A}\mathbf{X}_{t-1}||_F^2 \tag{6}$$

whose solution can be obtained by the singular value decomposition (SVD) of $\mathbf{O}_{t-1}$ according to Remark 2 [51], [52], [53], [54].

Remark 2: The feature matrix $\mathbf{X}$ is estimated by simply calculating the SVD of $\mathbf{O}$ when the noise $\mathbf{N}$ is independent identically distributed. The noise uniformly increases singular values (SVs) in the direction of each eigenvector without changing the order of these SVs and even the estimation of the subspace. The spectral subspace is approximately spanned by k singular vectors of $\mathbf{O}$ corresponding to the first-k SVs

$$\mathbf{A}_t=\mathbf{U}_t^{\mathbf{o}}(:,1:k) \tag{7}$$

where $\mathbf{U}_t^{\mathbf{o}}\in\mathbb{R}^{z\times z}$ is an orthogonal matrix and $[\mathbf{U}_t^{\mathbf{o}},\mathbf{S}_t^{\mathbf{o}},\mathbf{V}_t^{\mathbf{o}}]=\mathrm{SVD}(\mathbf{O}_t)$.

2) *Update* $\mathcal{W}_t$*:* Extract all their related items from (5)

$$\begin{aligned}\mathcal{W}&=\underset{\mathcal{W}}{\operatorname{argmin}}\ \gamma\sum_i||\mathrm{R}_i\mathcal{W}||_{w,*}+\frac{\mu}{2}||\mathcal{X}-\mathcal{W}+\mathcal{Y}_2/\mu||_F^2\\ &=\underset{\mathcal{W}}{\operatorname{argmin}}\ \gamma\sum_i||\mathrm{R}_i\mathcal{W}||_{w,*}+\frac{\mu}{2}||\mathcal{W}-(\mathcal{X}+\mathcal{Y}_2/\mu)||_F^2.\end{aligned} \tag{8}$$

Problem (8) in the form of tensor unfolding along mode-2 is

$$(\mathbf{W}_i)_t=\underset{\mathbf{W}_i}{\operatorname{argmin}}||\mathbf{W}_i||_{w,*}+\frac{\mu}{2\gamma}||\mathbf{W}-(\mathbf{X}_{t-1}+(\mathbf{Y}_2)_{t-1}/\mu)||_F^2 \tag{9}$$

which can be solved by the closed-form solution [55]

$$(\mathbf{W}_i)_t=\mathbf{U}_t^{\mathbf{w}}\hat{\mathbf{S}}_t^{\mathbf{w}}\left(\mathbf{V}_t^{\mathbf{w}}\right)^{\mathrm{T}} \tag{10}$$

where $[\mathbf{U}_t^{\mathbf{w}},\mathbf{S}_t^{\mathbf{w}},\mathbf{V}_t^{\mathbf{w}}]=\mathrm{SVD}(\mathbf{X}_{t-1}+(\mathbf{Y}_2)_{t-1}/\mu)$; $\mathbf{S}_t^{\mathbf{w}}$ is a diagonal matrix and its ith diagonal elements are denoted by σ_i. The ith SV $\hat{\sigma}_i$ of $\hat{\mathbf{S}}_t^{\mathbf{w}}$ can be computed by

$$\hat{\sigma}_i=\begin{cases}0, & \text{if } c_2<0\\ \dfrac{c_1+\sqrt{c_2}}{2}, & \text{if } c_2\geq 0\end{cases} \tag{11}$$

where $c_1=\sigma_i-\varepsilon$, $c_2=(\sigma_i+\varepsilon)^2-4C$, and C is positive.

3) *Update* $\mathbf{L}_t$*:* The $\mathbf{L}$-related subproblem is equivalent to the following constrained minimization problem:

$$\min_{\mathbf{L}}\alpha||\mathbf{L}||_*+\frac{1}{2}||\mathbf{X}-\mathbf{D}\mathbf{L}-\mathbf{E}||_F^2. \tag{12}$$

The quadratic term f in (7) is linearized at previous iteration $\mathbf{L}_{t-1}$. With the addition of a proximal term, the iteration of $\mathbf{L}_t$ goes as follows:

$$\begin{aligned}\mathbf{L}_t&=\underset{\mathbf{L}}{\operatorname{argmin}}\ \alpha||\mathbf{L}||_*+\frac{\eta}{2}||\mathbf{L}-\mathbf{L}_{t-1}||_F^2\\ &\quad+\left\langle\nabla_{\mathbf{L}}f(\mathbf{L}_{t-1},\mathbf{X}_{t-1},\mathbf{E}_{t-1},\beta_{t-1}),\mathbf{L}-\mathbf{L}_{t-1}\right\rangle\\ &=\underset{\mathbf{L}}{\operatorname{argmin}}\ \alpha||\mathbf{L}||_*\\ &\quad+\frac{\eta}{2}||\mathbf{L}-\mathbf{L}_{t-1}+[-\mathbf{D}^{\mathrm{T}}(\mathbf{X}_{t-1}-\mathbf{D}\mathbf{L}_{t-1}-\mathbf{E}_{t-1})]/\eta||_F^2\end{aligned} \tag{13}$$

where the quadratic term $\nabla_{\mathbf{L}}f$ is the partial differential of f with respect to $\mathbf{L}$ and $\eta=||\mathbf{D}||_F^2$.

4) *Update* $\mathbf{E}_t$*:* The $\mathbf{E}$-related subproblem is rewritten as

$$\min_{\mathbf{E}}\lambda||\mathbf{E}||_1+\frac{1}{2}||\mathbf{X}-\mathbf{D}\mathbf{L}-\mathbf{E}||_F^2. \tag{14}$$

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TABLE I
AUC VALUES OF ALL COMPARED ALGORITHMS FOR FOUR HS DATASETS UNDER FOUR NOISE CASES

data sets	Noise case	global RX [8]	DWT RX [57]	CKRX [14]	RPCA [24]	LRSAR [27]	GTVLRR [29]	LSMAD [25]	DSR-ADNR
Airport	Case 1	0.8654	0.8237	0.8627	0.7695	0.8517	0.8654	0.9051	**0.9501**
	Case 2	0.7293	0.7726	0.8250	0.6863	0.7897	0.8298	0.8277	**0.9406**
	Case 3	0.6694	0.7286	0.7756	0.6268	0.6554	0.8039	0.8190	**0.9270**
	Case 4	0.6467	0.7084	0.7359	0.6150	0.6896	0.7627	0.8202	**0.9464**
San Diego	Case 1	0.8614	0.9312	0.9552	0.8827	0.9562	0.9704	0.9763	**0.9903**
	Case 2	0.8102	0.9126	0.9380	0.7975	0.9218	0.9444	0.9707	**0.9894**
	Case 3	0.7917	0.8277	0.8532	0.7835	0.8984	0.9388	0.9428	**0.9887**
	Case 4	0.6039	0.6823	0.8042	0.5926	0.7846	0.8323	0.8901	**0.9897**
AIR-HI-Vehicle-I	Case 1	0.8443	0.8678	0.8818	0.8080	0.7691	0.9234	0.9235	**0.9563**
	Case 2	0.7555	0.8075	0.8551	0.7026	0.6276	0.8743	0.8705	**0.9460**
	Case 3	0.7518	0.8046	0.7653	0.6842	0.5680	0.7534	0.8438	**0.9367**
	Case 4	0.5530	0.6175	0.5425	0.5153	0.5093	0.5725	0.7935	**0.9363**
AIR-HI-Vehicle-II	Case 1	0.7937	0.8373	0.8788	0.7691	0.6606	0.8499	0.8939	**0.9341**
	Case 2	0.7867	0.8212	0.8295	0.7581	0.6042	0.8221	0.8711	**0.9335**
	Case 3	0.7222	0.7760	0.7896	0.6888	0.5352	0.6740	0.8712	**0.9181**
	Case 4	0.5073	0.5786	0.4002	0.4663	0.4588	0.5099	0.7826	**0.9337**

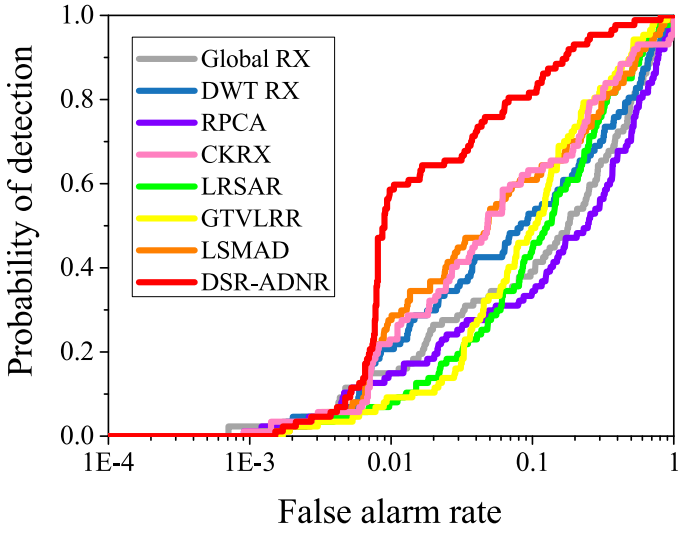


Fig. 4. ROC curves obtained by different methods for the Airport dataset.

The operation $\Delta_\varepsilon[x] = \text{sgn}(x).*\max(|x|-\varepsilon, 0)$ is used to obtain the solution of (15)

$$\mathbf{E}_t = \Delta_\lambda[\mathbf{X}_{t-1} - \mathbf{D}\mathbf{L}_t]. \tag{15}$$

5) Update $\mathbf{X}_t$ by minimizing the matrix form of $\mathbf{X}$-related subproblem as follows:

$$F(\mathbf{X}) = \frac{1}{2}||\mathbf{O} - \mathbf{A}\mathbf{X}||_F^2 + \frac{\mu}{2}\left(||\mathbf{X} - \mathbf{D}\mathbf{L} - \mathbf{E} + \mathbf{Y}_1/\mu||_F^2 + ||\mathbf{X} - \mathbf{W} + \mathbf{Y}_2/\mu||_F^2\right). \tag{16}$$

Set the derivative of (16) with respect to $\mathbf{X}$ to zero, and obtain the following equation:

$$\mathbf{X}_t = \frac{\mathbf{A}_t^{\mathrm{T}}\mathbf{O}_t + \mu\mathbf{D}\mathbf{L}_t + \mu\mathbf{E}_t - (\mathbf{Y}_1)_t + \mu\mathbf{W}_t - (\mathbf{Y}_2)_t}{1+2\mu}. \tag{17}$$

1) Iteration Refinement: Iterative regularization [56] is utilized to refine the observation HS image $\mathbf{O}_t$ in each iteration

$$\mathbf{O}_{t+1} = \Theta\mathbf{H}_t + (1-\Theta)\mathbf{O} \tag{18}$$

where Θ ($0 \le \Theta \le 1$) is the iterative regularization parameter. $\mathbf{H}_t = \mathbf{A}_t\mathbf{X}_t$ is the denoised result, and $\mathbf{O}$ is the noisy HS image. Previous methods fixed $\mathbf{A}$ and regarded $\mathbf{A}$ as a preprocessing step due to a fixed input $\mathbf{O}$. $\mathbf{A}_t$ is calculated from the lower noise variance of the intermediate value $\mathbf{O}_{t+1}$.

DSR-ADNR aims at the final interpretation of AD, and thus, we consider the convergence condition as follows:

$$\max\left\{\begin{array}{l} ||\mathbf{E}_t - \mathbf{E}_{t-1}||_\infty \\ ||\mathbf{L}_t - \mathbf{L}_{t-1}||_\infty \\ ||\mathbf{X}_t - \mathbf{X}_{t-1}||_\infty \\ ||\mathbf{X}_t - \mathbf{D}*\mathbf{L}_t - \mathbf{E}_t||_\infty \end{array}\right\} \le \epsilon. \tag{19}$$

Summarizing the aforementioned descriptions, the designed algorithm for DSR-ADNR is provided in Algorithm 1.

2) Computational Complexity: The computational complexity of the proposed DSR-ADNR is discussed as follows.

1) The main cost of updating (7) lies in the SVD operator, which requires about $z(\text{hv})^2 + z^3$ operations.
2) Updating (11) costs about $k(\text{hv})^2 + k^3$ operations.
3) The computation of solving (13) requires about $2\text{kdhv} + 2\text{khv} + 2\text{dhv} + k(\text{hv})^2 + k^3$ operations.
4) Solving the subproblem (15) takes about $\text{kdhv} + 2\text{khv}$ operations in each iteration.
5) Updating (17) needs about $\text{kzhv} + \text{kdhv} + 5\text{khv}$ operations.

The overall computational complexity of Algorithm 1 is $z(\text{hv})^2 + z^3 + 2k(\text{hv})^2 + 2k^3 + 4\text{kdhv} + 9\text{khv} + 2\text{dhv} + \text{kzhv}$. Generally, $k, d \ll z \ll \text{hv}$, and the final order of complexity for our DSR-ADNR is $O(z(\text{hv})^2)$. In addition, Section III-C2 provides an empirical running time comparison.

Remark 3: In [50], the alternating direction method of multipliers (ADMM) is generalized to gain LADMAP by linearizing the quadratic penalty term. The convergence of LADMAP has been proven for convex optimization problems with two blocks. It is worth noting that the separable and independent variables are treated as a single block. Nevertheless, unlike the four-block convex function (1) of DSR-ADNR,

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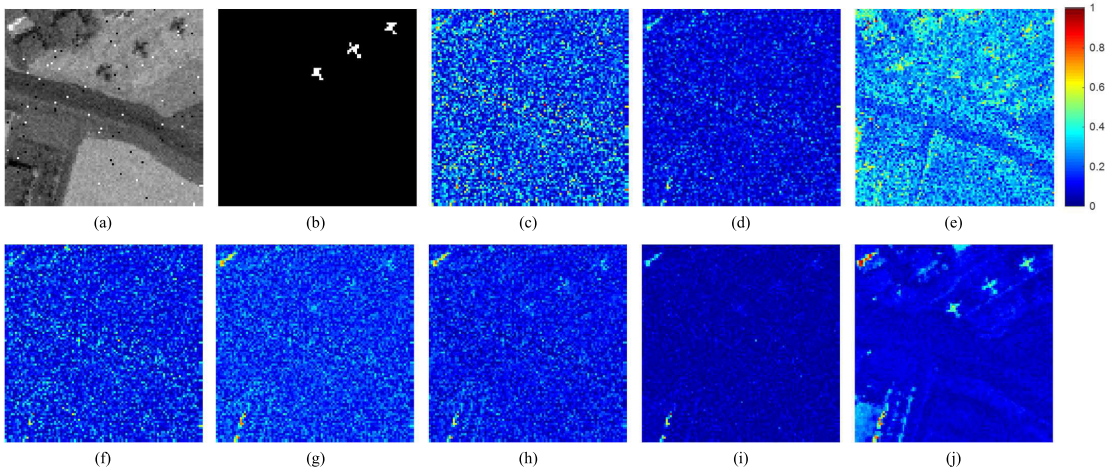


Fig. 5. Detection results of all compared methods: (a) noisy San Diego dataset under case 4, (b) reference map, (c) global RX, (d) DWT RX, (e) CKRX, (f) RPCA, (g) LRSAR, (h) GTVLRR, (i) LSMAD, and (j) DSR-ADNR.

Algorithm 1 DSR-ADNR Algorithm

Input: Input matrix with noise $\mathbf{O}_1 = \mathbf{O}$, a conducted dictionary $\mathbf{D}$, parameter γ, β, λ, μ, T, k, $\mu_{max} = 10^{10}$, $\rho_0 = 1.1$

while $0 \le t \le T$ or meet the convergence conditions

update $\mathbf{A}_t$ by calculating the singular value decomposition (SVD) of $\mathbf{O}_{t-1}$, e.g., $[\mathbf{U}_t^{\mathbf{o}}, \mathbf{S}_t^{\mathbf{o}}, \mathbf{V}_t^{\mathbf{o}}] = \mathrm{SVD}(\mathbf{O}_t)$, $\mathbf{A}_t = \mathbf{U}_t^{\mathbf{o}}(:, 1:k)$,

update $(\mathbf{W}_i)_t$ by solving Eq. (10): $(\mathbf{W}_i)_t = \mathbf{U}_t^{\mathbf{w}} \hat{\mathbf{S}}_t^{\mathbf{w}} (\mathbf{V}_t^{\mathbf{w}})^{\mathrm{T}}$,

update $\mathbf{L}_t$ by solving Eq. (13): $= \underset{\mathbf{L}}{\operatorname{argmin}}\, \alpha||\mathbf{L}||_* + \frac{\eta}{2}||\mathbf{L} - \mathbf{L}_{t-1} + [-\mathbf{D}^{\mathrm{T}}(\mathbf{X}_{t-1} - \mathbf{D}\mathbf{L}_{t-1} - \mathbf{E}_{t-1})]/\eta||_F^2$,

update $\mathbf{E}_t$ by solving Eq. (15):

$\mathbf{E}_t = \Delta_\lambda[\mathbf{X}_{t-1} - \mathbf{D}\mathbf{L}_t]$,

update $\mathbf{X}$ by solving Eq. (17):

$\mathbf{X}_t = \frac{\mathbf{A}_t^{\mathrm{T}}\mathbf{O}_t + \mu\mathbf{D}\mathbf{L}_t + \mu\mathbf{E}_t - (\mathbf{Y}_1)_t + \mu\mathbf{W}_t - (\mathbf{Y}_2)_t}{1+2\mu}$,

update the multipliers $(\mathbf{Y}_1)_t$ and $(\mathcal{Y}_2)_t$:

$(\mathbf{Y}_1)_t = (\mathbf{Y}_1)_{t-1} + \mu(\mathbf{X}_t - \mathbf{D}\mathbf{L}_t - \mathbf{E}_t)$,

$(\mathcal{Y}_2)_t = (\mathcal{Y}_2)_{t-1} + \mu(\mathcal{X}_t - \mathcal{W}_t)$,

$\mathbf{O}_{t+1} = \Theta\mathbf{H}_t + (1-\Theta)\mathbf{O}$,

Update μ as: $\mu = \min(\mu_{\max}, \rho\mu)$, where

$$\rho = \begin{cases} \rho_0, & \text{if } \mu \max\left(\sqrt{\eta}\|\mathbf{L}_{t+1} - \mathbf{L}_t\|_F, \|\mathbf{E}_{t+1} - \mathbf{E}_t\|_F, \right. \\ & \left. \|\mathbf{X}_{t+1} - \mathbf{X}_t\|_F\right)/\|\mathbf{X}\|_F \le \varepsilon_2 \\ 1, & \text{otherwise} \end{cases}$$

$t = t + 1$,

end while

Output: Anomaly detection map and denoised result $\mathbf{H} = \mathbf{A}\mathbf{X}$.

the extended version of the ADMM is not necessarily convergent even with the strong convex assumption [Theorem 4.1] [57]. As a result, we have thoroughly discussed the convergence performance of the DSR-ADNR framework. A further illustration of the convergence analysis is presented in Section III-C1.

III. Experimental Results and Analysis

In this section, extensive experiments are conducted to evaluate the performance of our proposed method. Several HS AD algorithms are compared with DSR-ADNR, including classic global RX [8], discrete wavelet transform (DWT RX) [58], CKRX [14], RPCA [23], [24], LRSAR [27], LSMAD [25], and graph and TV regularized LRR (GTVLRR) [29]. The parameters of these approaches are either adjusted or set to the values described in their respective works. Except for the visual comparison of denoised results, we adopt two quality evaluation indices: peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM). PSNR and SSIM calculate the similarity between the denoised result and the ground truth based on the mean squared error and structural similarity, respectively. These assessments are used for each spectral band. MPSNR and MSSIM are cast by averaging the PSNR and SSIM values, respectively, to illustrate the denoising precision along the spatial direction

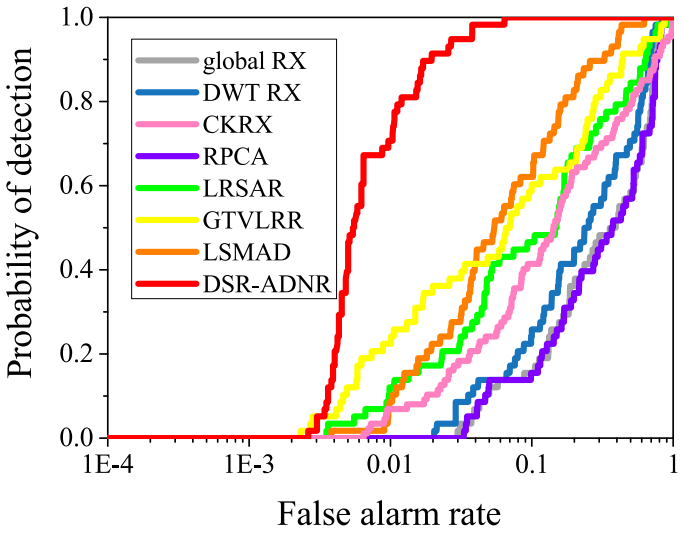


Fig. 6. ROC curves obtained by different methods for the San Diego dataset.

$$\mathrm{MPSNR} = \frac{1}{z}\sum_{k=1}^{z} \mathrm{PSNR}(\mathcal{H}(:,:,k), \overline{\mathcal{H}}(:,:,k)) \tag{20}$$

$$\mathrm{MSSIM} = \frac{1}{z}\sum_{k=1}^{z} \mathrm{SSIM}(\mathcal{H}(:,:,k), \overline{\mathcal{H}}(:,:,k)) \tag{21}$$

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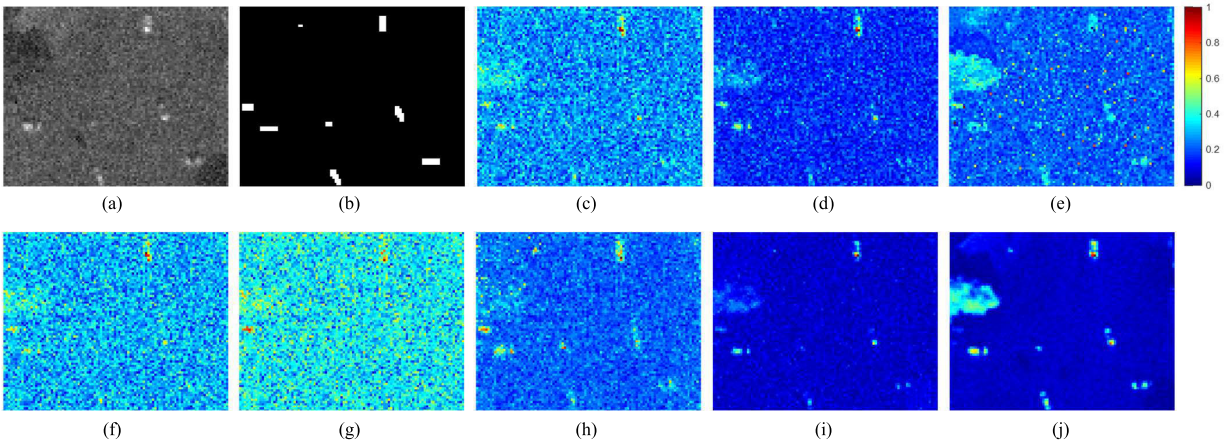


Fig. 7. Detection results of all compared methods: (a) noisy AIR-HI-Vehicle-I dataset under case 2, (b) reference map, (c) global RX, (d) DWT RX, (e) CKRX, (f) RPCA, (g) LRSAR, (h) GTVLRR, (i) LSMAD, and (j) DSR-ADNR.

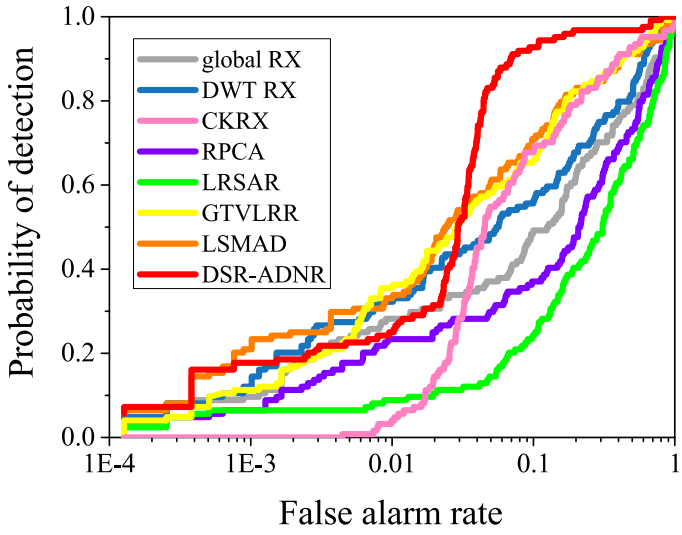


Fig. 8. ROC curves obtained by different methods for the AIR-HI-Vehicle-I dataset.

where $\mathcal{H}(:,:,k)$ and $\overline{\mathcal{H}}(:,:,k)$ denote the ith band of the reference and denoised images, and $\mathbf{H}_{(3)}(k,:)$ and $\overline{\mathbf{H}}_{(3)}(k,:)$ are the ith spectral signatures of two images.

For AD, color detection maps are provided in all experiments, with the colors ranging from blue to red. The receiver operating characteristic (ROC) curve [59] is given to reflect the compromise between the probability of detection (PD) and FAR. The area under the ROC curve, AUC [60], is presented to compare different algorithms quantitatively.

A. Simulated HS Data Experiments

1) HS Datasets' Description:

a) San Diego scene: The San Diego dataset was collected by the airborne visible/infrared imaging spectrometer (AVIRIS) in San Diego, CA, USA. This dataset has a spatial resolution of 3.5 m with a spectral range between 370 and 2510 nm. This scene consists of 400 × 400 pixels and 224 bands, while a subimage of 100 × 100 × 189 is usually used for experimental analysis. The background in this scene refers to various types of grass, roofs, roads, and so on. Three manual airplanes, on the other side, are considered anomalies.

b) Airport scene: This sample image of size 100 × 100 is extracted from the data downloaded from the AVIRIS website.[1] This dataset was acquired by AVRIS in Los Angeles with a spatial resolution of 7.1 m. In this scene, two aircraft refer to outliers.

c) AIR-HI dataset: This aerial dataset was acquired by the visible and near-infrared imaging spectrometer designed by the Shanghai Institute of Technical Physics, Chinese Academy of Sciences, Xiong'an New Area, China. The flight height of this image is 2000 m, and the spatial resolution of the collected imagery is 0.525 m per pixel. The whole image contains 3751 × 2746 pixels and 253 bands in the 400–1000-nm spectral range. In this Aerospace Information Research Institute HS image (AIR-HI), several vehicles are considered anomalies in the urban scene. Hence, in this experiment, two split subimages are named AIR-HI-Vehicle-I and AIR-HI-Vehicle-II, with the size of 80 × 100 and 80 × 120, respectively. In the AIR-HI-Vehicle-I dataset, the number of anomalies is 124, with a 1.55% proportion. In the second one, anomalous pixels contain 166 pixels, accounting for 1.73% of the total. These anomalies in the AIR-HI dataset are manually labeled by the Quantum GIS (QGIS) and characterized based on four features [61], [62]: no prior knowledge, low probability of occurrence, small spatial size, and insignificance in spectral statistics.

d) Experimental setting: We add additive noise with different variances into three datasets to test the denoising and detection performances of the proposed DSR-ADNR. For the first three noise cases, the Gaussian noise with $\sigma/255 = \{10, 20, 30\}/255$ is used to simulate the noisy images.

Noise Case 4: Zero-mean Gaussian noise with $\sigma/255 = 10/255$, and 1% salt and pepper noise.

[1] http://aviris.jpl.nasa.gov/

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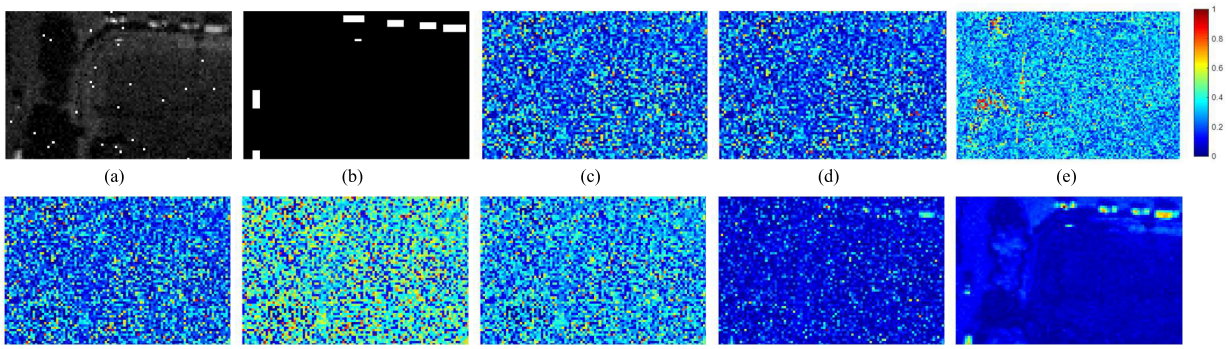


Fig. 9. Detection results of all compared methods: (a) noisy AIR-HI-Vehicle-II dataset under case 4, (b) reference map, (c) global RX, (d) DWT RX, (e) CKRX, (f) RPCA, (g) LRSAR, (h) GTVLRR, (i) LSMAD, and (j) DSR-ADNR.

TABLE II

DENOISED PERFORMANCES BY DIFFERENT DENOISING METHODS FOR FOUR HS DATASETS UNDER FOUR NOISE CASES

HS data sets	Noise case	Index	Noisy	Wavelet [45]	BM3D [46]	BM4D [47]	DSR-ADNR
Airport	Case 1	MPSNR	28.13dB	30.02dB	32.35dB	36.80dB	**40.58dB**
		MSSIM	0.8000	0.8577	0.9225	0.9680	**0.9836**
	Case 2	MPSNR	22.11dB	25.05dB	28.92dB	33.05dB	**36.73dB**
		MSSIM	0.5533	0.6711	0.8447	0.9290	**0.9634**
	Case 3	MPSNR	18.59dB	22.65dB	27.11dB	31.85dB	**33.84dB**
		MSSIM	0.3884	0.5449	0.7782	0.9131	**0.9306**
	Case 4	MPSNR	23.41dB	25.26dB	25.32dB	33.92dB	**37.40dB**
		MSSIM	0.6987	0.7589	0.8074	0.9405	**0.9698**
San Diego	Case 1	MPSNR	28.13dB	30.43dB	33.56dB	37.55dB	**40.21dB**
		MSSIM	0.6837	0.7863	0.9113	0.9576	**0.9746**
	Case 2	MPSNR	22.11dB	25.42dB	30.02dB	35.20dB	**36.74dB**
		MSSIM	0.4151	0.5652	0.8376	0.9259	**0.9350**
	Case 3	MPSNR	18.60dB	23.01dB	28.32dB	32.35dB	**33.91dB**
		MSSIM	0.2742	0.4446	0.7857	0.8671	**0.8804**
	Case 4	MPSNR	23.59dB	25.86dB	29.08dB	34.25dB	**37.65dB**
		MSSIM	0.5762	0.7026	0.8011	0.9142	**0.9491**
AIR-HI-Vehicle-I	Case 1	MPSNR	28.13dB	28.94dB	37.00dB	38.24dB	**41.34dB**
		MSSIM	0.4955	0.7189	0.9031	0.9191	**0.9428**
	Case 2	MPSNR	22.11dB	26.09dB	32.89dB	33.32dB	**35.52dB**
		MSSIM	0.2364	0.4770	0.7870	0.8199	**0.8301**
	Case 3	MPSNR	18.59dB	24.18dB	31.28dB	32.05dB	**33.05dB**
		MSSIM	0.1403	0.6476	0.8009	0.8140	**0.8298**
	Case 4	MPSNR	22.88dB	25.11dB	26.74dB	33.61dB	**38.01dB**
		MSSIM	0.4016	0.6787	0.7651	0.8278	**0.8849**
AIR-HI-Vehicle-II	Case 1	MPSNR	28.13dB	31.13dB	34.82dB	39.74dB	**42.88dB**
		MSSIM	0.6005	0.7472	0.9046	0.9589	**0.9734**
	Case 2	MPSNR	22.11dB	26.52dB	31.95dB	34.11dB	**37.36dB**
		MSSIM	0.3218	0.5256	0.8561	0.8674	**0.9114**
	Case 3	MPSNR	18.59dB	23.73dB	30.02dB	32.15dB	**33.70dB**
		MSSIM	0.1975	0.3739	0.8081	0.8237	**0.8237**
	Case 4	MPSNR	22.82dB	25.42dB	28.85dB	32.62dB	**38.09dB**
		MSSIM	0.5106	0.6921	0.7688	0.8678	**0.9200**

TABLE III

AUC VALUES OF ALL COMPARED ALGORITHMS FOR TWO REAL HS DATASETS

HS data sets	global RX [8]	DWT RX [57]	CKRX [14]	RPCA [24]	LRSAR [27]	GTVLRR [29]	LSMAD [25]	DSR-ADNR
Real Airport-II	0.9287	0.9331	0.9266	0.9275	0.9212	0.9293	0.9379	**0.9592**
Real San Diego-II	0.9534	0.9743	0.9671	0.9484	0.9238	0.9588	0.9779	**0.9915**

2) Detection Performances:

a) Airport: In Fig. 3(a), the noisy Airport image is corrupted under noise case 2. Fig. 3(b) shows the corresponding reference detection map. The Gaussian noise has a more negative influence on CKRX, LRSAR, and GTVLRR than global RX, DWT RX, and RPCA. This type of noise widely distributes the whole image and makes higher false alarms in the results of global RX, DWT RX, and RPCA. LSMAD outperforms the first six approaches, but the edges of flights are destroyed by the Gaussian noise. As displayed in Fig. 3(j),

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TABLE IV
TIME CONSUMPTION (SECONDS) OF DIFFERENT METHODS ON DIFFERENT DATASETS

HS data sets	global RX [8]	DWT RX [57]	CKRX [14]	RPCA [24]	LRSAR [27]	GTVLRR [29]	LSMAD [25]	DSR-ADNR
Airport	0.06s	4.13s	0.58s	5.07s	17.42s	145.66s	6.70s	15.68s
San Diego-II	0.06s	3.60s	0.60s	2.64s	15.11s	128.69s	6.08s	13.89s
AIR-HI-Vehicle-I	0.06s	3.74s	0.62s	7.09s	25.04s	164.18s	7.35s	22.16s
AIR-HI-Vehicle-II	0.06s	5.26s	0.62s	9.43s	22.64s	176.61s	9.21s	18.89s
Real Airport-II	0.05s	3.92s	0.66s	4.13s	19.32s	137.04s	6.99s	16.40s
Real San Diego-II	0.05s	4.52s	0.58s	4.88s	21.96s	143.74s	8.23s	17.47s

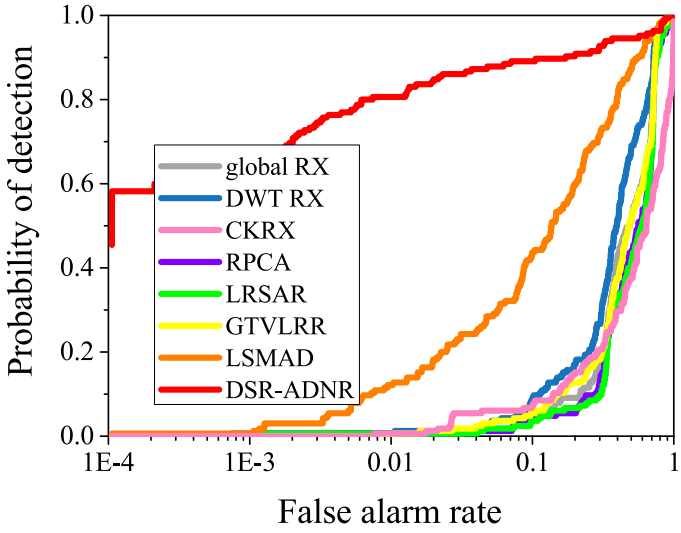


Fig. 10. ROC curves obtained by different methods for the AIR-HI-Vehicle-II dataset.

DSR-ADNR achieves the most accurate detection map with less noise among all competing algorithms.

Fig. 4 shows the ROC curves, which are plotted by PDs and FARs to quantitatively assess the performances of DSR-ADNR and the other methods. DSR-ADNR in red performs better than the others. For example, the PD of DSR-ADNR reaches 1.0 even when FAR is 0.6, yet most of the others are lower than about 0.85. Table I reports the AUC values of different anomaly detectors for four HS images under four noise cases. Consistent with the visual comparison, three LR-based approaches, e.g., LRSAR, LSMAD, CKRX, and GTVLRR, gain higher scores than the other two RX-based ones and RPCA. Considering the Gaussian noise in the degradation model, LSMAD with underlines produces the second-best detection performance than the others. The proposed model surpasses LSMAD by more than 0.05 under all noise cases.

b) San Diego: In terms of visual quality, Fig. 5 shows the detection results for the San Diego dataset under noise case 4. Fig. 5(a) and (b) provides the noisy image and reference detection map. Due to the negative influence of noise, higher false alarms exist in the results of global RX, DWT RX, and RPCA than in the three LR-based methods. Three flights can be slightly observed in the detection results of CKRX, LRSAR, LSMAD, and GTVLRR. The proposed DSR-ADNR still suppresses most of the mixed noise, which benefits obvious anomalies detected in Fig. 5(j).

For illustrative proposes, the ROC curves of DSR-ADNR and the other competing approaches are displayed in Fig. 6. DSR-ADNR achieves the highest PDs among all algorithms for most FARs, even reaching the peak when FAR is 0.06. The AUC values are provided in the third row of Table I. In accordance with the visual comparison, DSR-ADNR exhibits the highest scores as expected among all detectors, even approximately 0.9900. LSMAD with underlines gains better detection performances than the other two LR-based algorithms since the Gaussian noise is modeled into the optimization problem. The performances of detected results by all methods worsen with the heavier noise cases, whereas DSR-ADNR suffers slower decays.

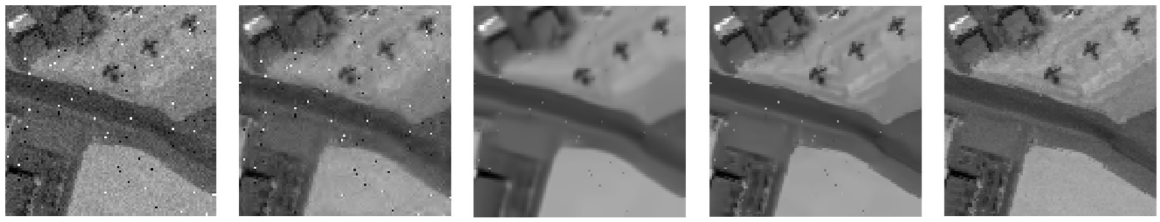
c) AIR-HI-Vehicle-I: In this dataset, several vehicles exist at the crossroad. Trees and one crossroad refer to the background. In terms of visual appeal, global RX, RPCA, and LRSAR fail to detect anomalies with the interference of noise in Fig. 7. Although most detectors perform lower false alarms, more or fewer anomalies still are observed in the results of DWT RX, CKRX, LSMAD, and GTVLRR. Especially, the response of the noise in the result of LSMAD is weaker than DWT RX, CKRX, and GTVLRR. DSR-ADNR undoubtedly achieves the best result with the clearest edges and structure of vehicles among all compared methods.

Fig. 8 depicts the ROC curves of DSR-ADNR and the other compared approaches. DSR-ADNR first shows mediocre performance, yet its curve in red dramatically rises after 0.02 and arrives at the largest value. GTVLRR and LSMAD always have higher PD values than the other four detectors. As the fourth row of Table I reports, the performances of the first seven detectors suffer dramatic deterioration, while DSR-ADNR performs more robustly to serious noise cases than the others. Our method surpasses GTVLRR by about 0.093 under noise case 3 on the scene of urban-I.

d) AIR-HI-Vehicle-II: Regarding visual appeal, Fig. 9(a) presents the noisy HS image with Gaussian noise, and salt and pepper noise. Six vehicles refer to anomalies in Fig. 9(b). The background contains trees, grass, and roads. There is no doubt that global RX, DWT RX, CKRX, and RPCA hardly enable robustness to mixed noise. Even for GTVLRR, the TV term is useless to remove sparse noise. LSMAD only identifies two anomalies located at the top-right corner of Fig. 9(h). The anomalies in the result of DSR-ADNR present stronger responses than the other things on the scene of Urban-II.

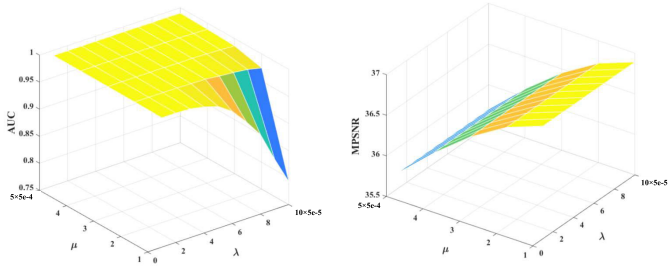
Fig. 10 illustrates the ROC curves obtained for the AIR-HI-Vehicle-I dataset. The proposed DSR-ADNR achieves the highest PD values for all FARs among all competing detectors.

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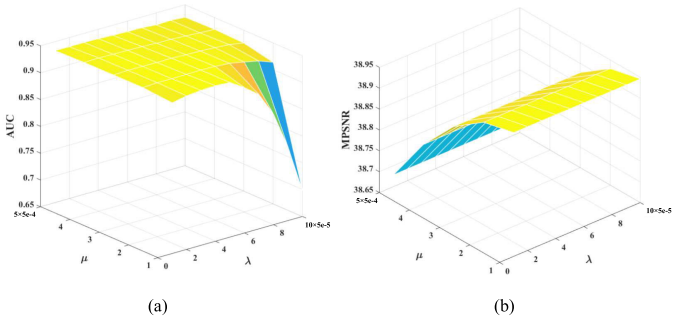
(a) (b) (c) (d) (e)

Fig. 11. Denoised results of all compared methods: (a) noisy San Diego dataset under case 4, (b) Wavelet, (c) BM3D, (d) BM4D, and (e) DSR-ADNR.



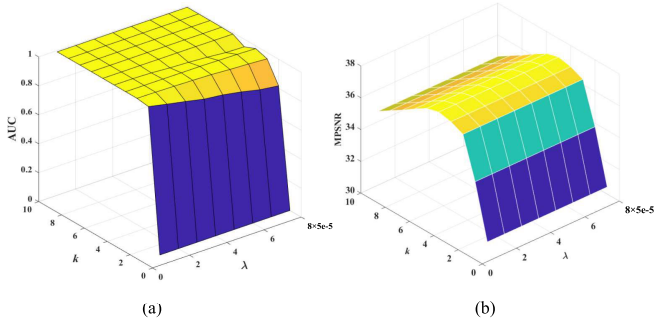
(a) (b)

Fig. 12. Parameter sensitivity analysis with respect to λ and μ for the San Diego dataset under case 2: (a) AUC values and (b) MPSNR values.



(a) (b)

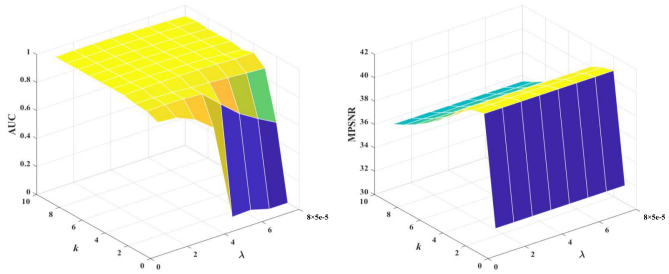
Fig. 13. Parameter sensitivity analysis with respect to λ and μ for the AIR-HI-Vehicle-I dataset under case 2: (a) AUC values and (b) MPSNR values.



(a) (b)

Fig. 14. Parameter sensitivity analysis in terms of λ and k for the San Diego dataset: (a) AUC values and (b) MPSNR values.

LSMAD in yellow outperforms the first six approaches, whose curves have similar and worse trends. Table I also gives the AUC values for the AIR-HI-Vehicle-I dataset. Except for LSMAD and DSR-ADNR, the others hardly cope with the mixed noise case and just obtain low AUC scores, even approximately 0.5. LSMAD detects the sparse noise with



(a) (b)

Fig. 15. Parameter sensitivity analysis in terms of λ and k for the AIR-HI-Vehicle-I dataset: (a) AUC values and (b) MPSNR values.

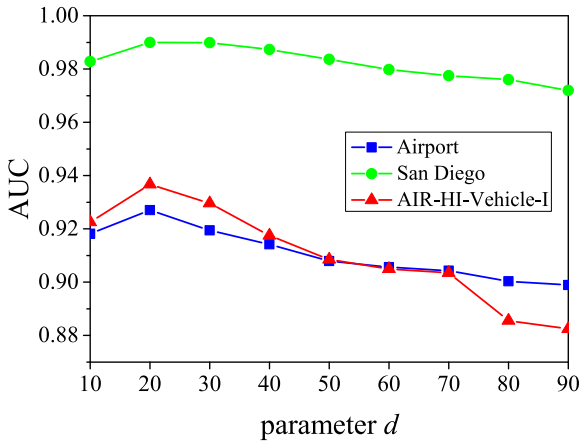


Fig. 16. AUC values of different numbers of dictionary atoms d for three datasets.

different locations wrongly and, hence, gains scores of less than 0.8 under case 4. Remarkably, our method achieves appealing performance even under a mixed noise case.

In summary, as detection maps of eight algorithms present, our DSR-ADNR always performs the best results under different noise cases, even for different sizes of datasets, different backgrounds, and anomalies. The detection performance of all methods degrades with increasing noise, whereas DSR-ADNR suffers from slower decays, which demonstrates the robustness of DSR-ADNR to different levels of noise cases.

3) Denoising Performance: We compare the denoising performance of the proposed DSR-ADNR with three other methods: Wavelet [45], BM3D [46], and BM4D [47]. The visual results of a single band in the San Diego dataset under case 4 are shown in Fig. 11. Wavelet is unable to entirely remove

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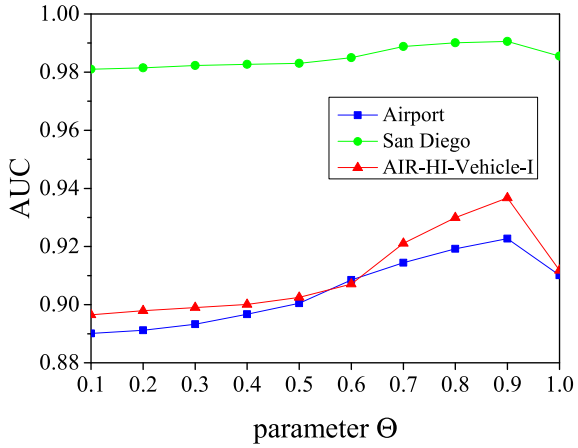


Fig. 17. AUC values of different parameters Θ for three datasets.

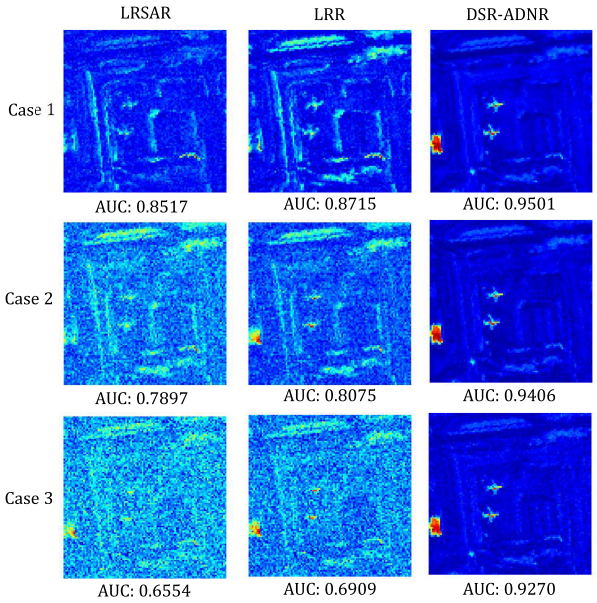


Fig. 18. Detection results obtained by LRSAR, LRR, and DSR-ADNR on the Airport dataset.

the sparse noise. BM3D and BM4D remove Gaussian noise, while some sparse noise still remains. BM3D losses more spatial information than BM4D. As observed, the proposed DSR-ADNR performs better than the other compared methods, maintaining clear texture information and sharp image edges. The quantitative results are presented in Table II, where the best ones are indicated in bold, and the second ones are underlined. Consistent with the visual observations, our proposed DSR-ADNR demonstrates superior performance compared to the other algorithms in terms of MPSNR and MSSIM. BM4D achieves the second-best MPSNR and SSIM scores.

4) Parameter Analysis: In the proposed DSR-ADNR, five parameters, λ, μ, k, the number of dictionary atoms, and Θ, are important to be determined. All the experimental analyses are conducted under case 2. Due to AD as the final goal of the joint optimization problem, we pay more attention to the relationship between λ and the other two parameters.

Parameter λ is crucial in controlling the strength of the sparsity term. Parameter μ is well related to the convergence speed in LADMAP iterations [63], [64]. These parameters λ and μ are simultaneously taken into consideration to investigate the AD and denoising performance gains in Figs. 12 and 13 for the San Diego dataset and the AIR-HI-Vehicle-I dataset. λ can be selected from the range of [5e-5, 5e-4] with an interval of 5e-5. μ varies from 5e-4 to 2.5e-3. The AUC curves show upward trends when λ decreases, and the denoised results present better performance when μ gets smaller.

Parameter k denotes the dimension of the spectral subspace and the impact of feature extraction for AD. Thus, this parameter is discussed with λ. k is chosen from [1, 10] with an interval of 1. As Fig. 14 and 15 reveal, for the San Diego dataset and the AIR-HI-Vehicle-I dataset, DSR-ADNR is sensitive to parameter k. When k increases from 1 to 5, the AUC and MPSNR curves rise sharply. After $k = 8$, the MPSNR performances fall off gradually. To sum up, the key parameter adjustment of DSR-ADNR is listed as follows: 1) choose μ as from the scope of [5e-4, 2.5e-3] with 5e-4 as step size; 2) the range of λ from [1e-4, 2e-4, 3e-4]; and 3) k is provided within [5, 8] with an interval of 1.

The number of atoms d plays a vital role in the dictionary construction of AD. Three datasets are taken as examples to evaluate the impact of different parameter settings on DSR-ADNR according to AUC. d is tuned from the range of [10, 90] with an interval of 10. Three curves rise at begin in Fig. 16. As the numbers of dictionary atoms increase, the detection performances of the proposed method gradually deteriorate. As the best AUC values are achieved when $d = 20$, we set $d = 20$ in all the experiments.

Iterative refinement has been proven to have a positive influence on noise removal [56]. Here, we emphatically investigate the impact of Θ on detection performances. Fig. 17 presents the AUC values achieved by DSR-ADNR with different parameters Θ. When Θ increases from 0.1 to 0.9, the AUCs show an upward trend, which indicates that $\mathbf{H}_t$ makes a positive effect on the detection performances. After peaking at $\Theta = 0.9$, all curves for three datasets fall off. Therefore, we set $\Theta = 0.9$.

5) Ablation Study: The contribution of the denoising term is investigated on the detection performance using the Airport dataset under noise cases 1–3. It is worth noting that the removal of the spectral-domain subspace component in our proposed DSR-ADNR model results in a degradation into an LRR model. As shown in Fig. 18, we compared the detection results of the LRSAR, LRR, and DSR-ADNR models. We observed that the detection accuracy of these methods decreases as the noise deteriorates. Interestingly, we found that the performance of LRASR is inferior to LRR, where LRASR adopts both LR and l_1 sparsity-inducing regularizations to capture the local structure of the background. The sparsity criterion assumes that each pixel covers a limited number of materials, while the low-rankness criterion arises from the fact that the number of distinct materials in a scene is typically much smaller than the number of spectral bands. However, the adoption of both criteria in LRSAR may be

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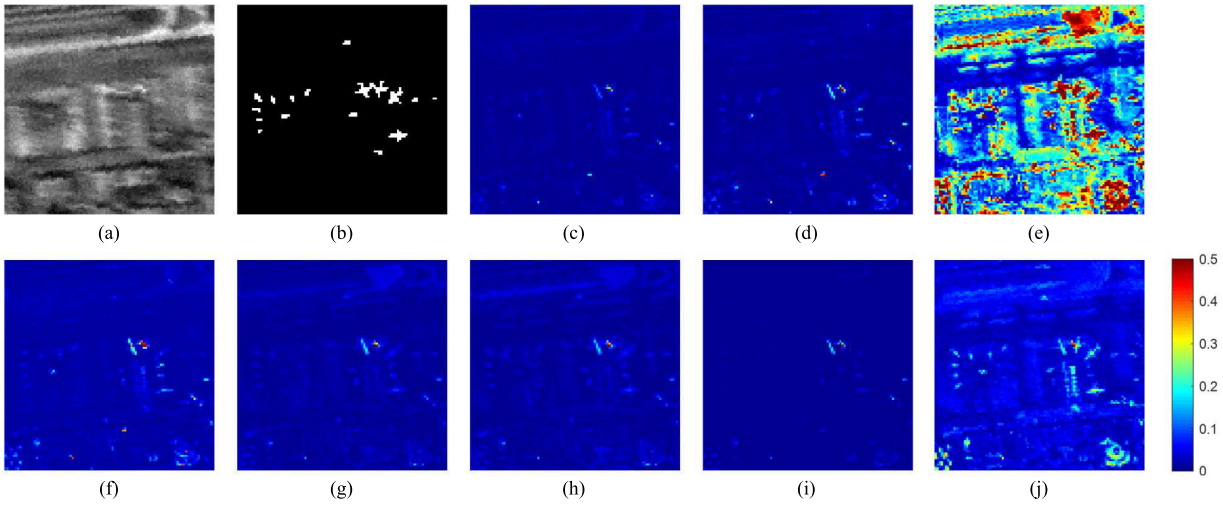


Fig. 19. Detection results of all compared methods: (a) real Airport-II dataset, (b) reference map, (c) global RX, (d) DWT RX, (e) CKRX, (f) RPCA, (g) LRSAR, (h) GTVLRR, (i) LSMAD, and (j) DSR-ADNR.

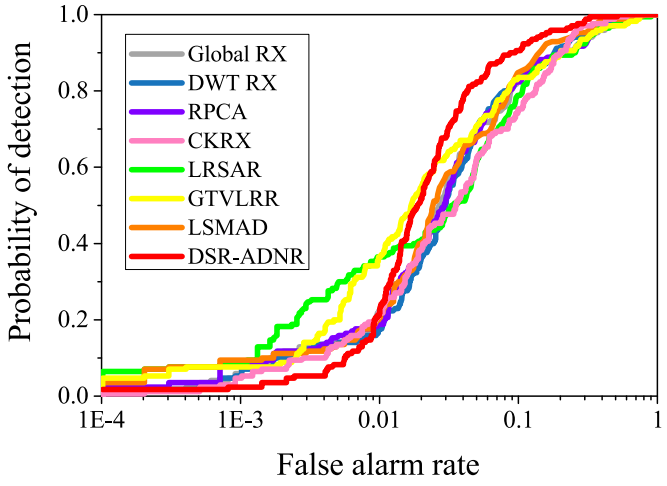


Fig. 20. ROC curves obtained by different methods for the real Airport-II dataset.

viewed as a partial repetition of the same underlying physical concept, and their combination could be considered somewhat redundant.

LRR's ability to handle noise is compromised without the assistance of the spectral-domain subspace component. In particular, when dealing with heavy noise (e.g., case 3), LRR may be incapable of accurately distinguishing the locations of two flights.

B. Real HS Data Experiments

In this section, we evaluate the proposed method on two real HS datasets captured at different scenes.

1) *Real Airport-II:* Fig. 19(a) and (b) depicts the noisy band and reference detection map of the real Airport-II dataset, respectively. This dataset has a size of 100 × 100 × 205 and is manually extracted from a large HS image downloaded from the AVIRIS Website.

2) *Real San Diego-II:* The image of the real San Diego-II dataset is of size 100 × 100 × 202, and some bands have been contaminated. The reference detection map of this dataset includes three anomalous flights.

1) Real Airport-II: Fig. 19 presents the detection maps generated by eight different approaches. Global RX and DWT RX produce obvious detection results on the right-hand side of Fig. 19(c) and (d). The edges of airports wrongly identified by CKRX lead to high FARs. LSAMD outperforms RPCA, LRSAR, and GTVLRR in suppressing the background and noise. DSR-ADNR shows a superior performance among all the approaches, which is consistent with the quantitative evaluation displayed in Table III. Due to its consideration of noise in mathematical modeling, LSMAD achieves the second-highest AUC score among the eight algorithms. The first six detectors gain similar AUCs, approximately 0.93. In Fig. 20, all curves exhibit similar trends of variation. However, the PDs of DSR-ADNR reach 1.0 when FAR = 0.5.

2) Real San Diego-II: In Fig. 21(a), the HS image is corrupted by the Gaussian noise. The detection of a single flight located at the top of figures is notably challenging for global RX, DWT RX, and RPCA. Furthermore, CKRX, LSRAR, and GTVLRR exhibit inadequate background suppression capabilities. Among the various techniques investigated, DSR-ADNR outperforms LSMAD in terms of false target identification. In Fig. 22, the PDs of DSR-ADNR are observed to be the highest when FAR is 0.08. To quantify these observations, we report the AUC scores of different algorithms in Table III. The proposed method yields superior performance, surpassing LSMAD, DWT RX, and CKRX by 1.36%, 1.72%, and 2.44%.

C. Discussion

1) Convergence Analysis: The numerical convergence is analyzed on four datasets under noise case 4. Fig. 23 and 24 depict the AUC and MPSNR values versus iterations,

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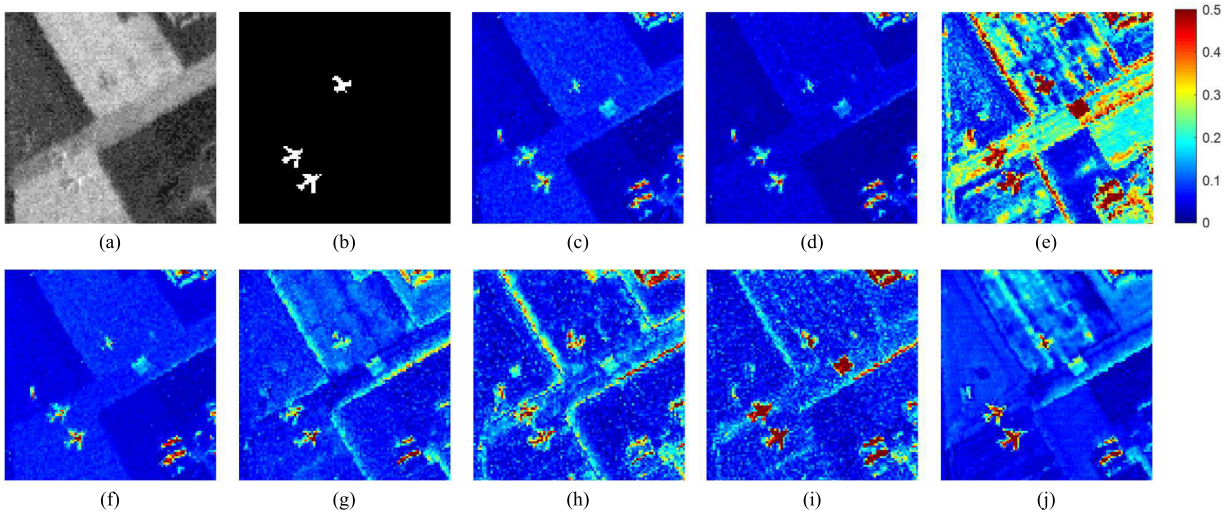


Fig. 21. Detection results of all compared methods: (a) real San Diego-II dataset, (b) reference map, (c) global RX, (d) DWT RX, (e) CKRX, (f) RPCA, (g) LRSAR, (h) GTVLRR, (i) LSMAD, and (j) DSR-ADNR.

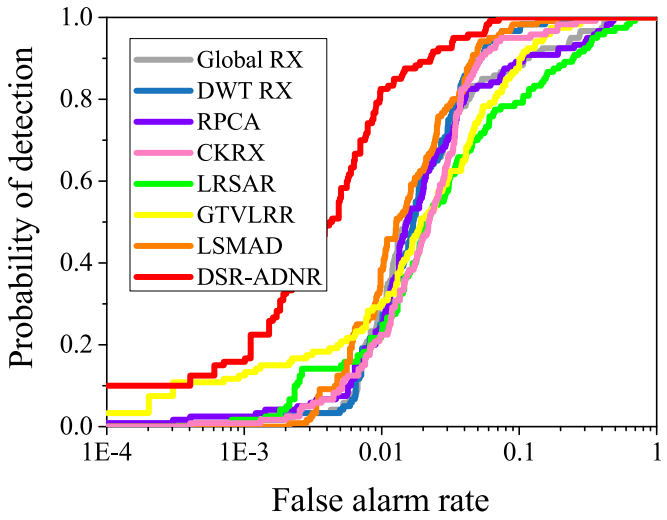


Fig. 22. ROC curves obtained by different methods for the real San Diego-II dataset.

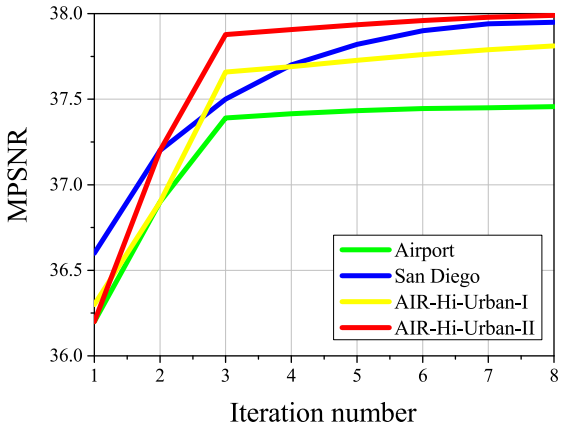


Fig. 24. MPSNR versus iterations on four datasets.

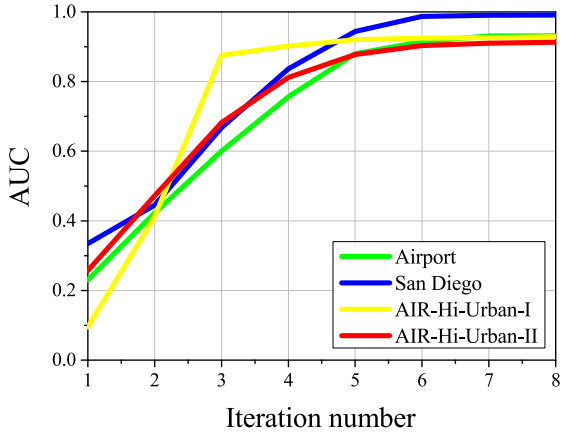


Fig. 23. AUC versus iterations on four datasets.

respectively. As the number of iterations increases, the AUC and MPSNR curves decrease monotonically at first and then remain stable, demonstrating the convergence of DSR-ADNR numerically.

2) Running Time: The running times of different approaches are listed in Table IV. All of the experiments are conducted in MATLAB 2018b on the same PC with an Intel i7 CPU at 2.90 GHz and 64 GB of memory. Global RX requires less time but obtains much worse detection accuracy than the proposed DSR-ADNR method on four datasets. DWT RX, RPCA, and LSMAD have similar running times, yet LSMAD is more robust to noise than the former. The running times of GTVLRR on all datasets are more than LRSAR and DSR-ADNR due to the fusion of the TV regularization. LRSAR requires a significant amount of computational time due to the calculation of SVD with the original image size. In contrast, DSR-ADNR conducts feature extraction via a spectral subspace and performs SVD on a low-dimensional feature matrix. This approach results in a reduced computational burden compared to LRSAR. The proposed DSR-ADNR jointly produces the detected objects and denoised results with appropriate running times.

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IV. Conclusion and Further Analysis

In this article, we proposed a unified framework called DSR-ADNR to complete joint HS AD and denoising tasks. The DSR-ADNR method utilizes a spectral space with a low-dimensional orthogonal basis to refine the distinctive feature of HS AD, which is more effective than directly using the original raw pixels. Based on the nonlocal self-similarity, noise removal also benefits the LR subspace representation of the feature matrix and, thus, improves the detection performance and computational efficiency. The LR property of FBPs is also enhanced due to the enlightened LRR of the background and anomalies. We developed the designed LADMAP to solve the DSR-ADNR optimization problem. Experimental results on simulated and real-world datasets demonstrate the superior performances of our proposed method. The denoised and detected results are output concurrently via the joint scheme. The proposed method represents a novel research direction for intelligent HS RS development, emphasizing the importance of practical considerations in HS imaging applications.

In the future, the tensor-based models will be investigated for jointly formulating and solving two tasks. Due to the 3-D inherent structure of HS images, tensor LRR has the potential ability to remove noise and represent the background and anomalies.

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